

Experience

Unity 3D Engine • 10+ years • PC/Mobile development	C/C++ • 20+ years • High perf systems, blockchains
Unreal Engine • 10+ years • Simulation development	Rust • 10+ years • Blockchain, AI, simulation
Computer Vision • 10+ years • AI/ML, rendering development	Python • 20+ years • Tools and pipeline engineering
GPGPU/CUDA • 10+ years • High performance computing	F#/C# • 10+ years • Apps, games and tools

Employment

Defense	AI Simulation Engineer	2018-
- Developing synthetic datasets for training ML models and virtual environments for training robots. - Developing scalable simulation systems for robot swarms using Unreal Engine, PX4, ROS (Robot Operating System) - Engineering autonomous AI agents for tactical mission planning, scenario generation, and CONOPS environments.		
Vertex Studio https://vr.x.group	Applied R&D	2016-
- Developing deep learning pipelines using PyTorch, real time pipelines using GStreamer and simulations using Unreal Engine, Unity 3D and custom engines. - Developing core blockchains for scaling protocol-owned economies for games, simulations and XR experiences. Smart contracts, DeFi, GameFi, market makers. - Maintaining custom Unreal Engine + NVidia PhysX for robotics applications.		
Simbotic https://simbotic.ai/	Open Source Developer	2019-
- Leading development of open source simulation and AI platform for bridging the reality gap with digital twins. - Working on real time pipelines for generating synthetic data for training AI models. - Integrating physics-based dynamics and computer vision into game engines. - Improving sample efficiency of reinforcement learning models with rich virtual environments with dense rewards. - Maintaining several docker containers for CUDA, ROS, GStreamer and TensorFlow. - Custom 3D engine development for vectorizing meta-layers for XR experiences.		
Civil Maps https://civilmaps.com/	Simulation Consultant	2017-2019
Velodyne https://velodynelidar.com/		
Galois https://galois.com/		
- Sensor characterization, customization and visualization tools. - Integration of Unreal Engine into development operations. - Massive autonomous swarming simulation architecture for DARPA, DoD and similar agencies.		
Red Pill VR http://redpillvr.com/	Systems Engineer & Technical Art	2016-2017
- Integrated real time deep learning inference for VR Music MMO game using TensorFlow, GStreamer and UE4. - Physics-based VR gameplay, UX interaction and full-body VR networking.		
Rawbots https://rawbots.github.io/	Indie Game Developer	2015-2016
Designed and develop Rawbots, a robot crafting sandbox game. Robot simulations developed with Bullet Physics.		
LucasArts & ILM https://www.lucasfilm.com/	AAA Game Developer	2005-2015
MunkyFun		
Beyond Games		
- Console game developer for Xbox, PlayStation, Nintendo, PC, iOS and Android. - Gameplay, physics and network developer for multiple titles, most notable Star Wars: The Force Unleashed. - Simulation tools for real-time special effects and pre-vis for several movies.		

Skills

C • C++ • Rust	ROS • PX4	Console • PC • Mobile	Git • Github • Gitlab
F# • C#	TensorFlow GPU	OpenGL • CUDA • GLSL	Win • Linux • Android • iOS
Python	GStreamer	THREE.js • A-Frame	Docker • AWS • GCP
Type/JavaScript	UnrealEngine • Unity3D	LLMs • Vision	TCP UDP WebRTC WebSockets
Erlang	Bullet Physics • PhysX	NVidia Xavier/Jetson	Make • CMake • Bazel